CHAPTER 3

RESEARCH METHODOLOGY

This chapter discusses the methodology used to develop **IronGrind:** A Web-Based Fitness Gym Membership and Operations Management System for Iron Grind Fitness Gym in Purok Mangga, Sulangon, Dapitan City, Zamboanga del Norte. It describes the development process, system requirements, design, data organization, and evaluation procedures. The system is intended to organize membership applications and records, membership periods and renewals, staff-recorded attendance, products, sales, and reports. Its development is organized using Agile, with work planned and delivered incrementally. Proposed evaluation activities are distinguished from results that have not yet been collected.

System Development Methodology

Agile Methodology

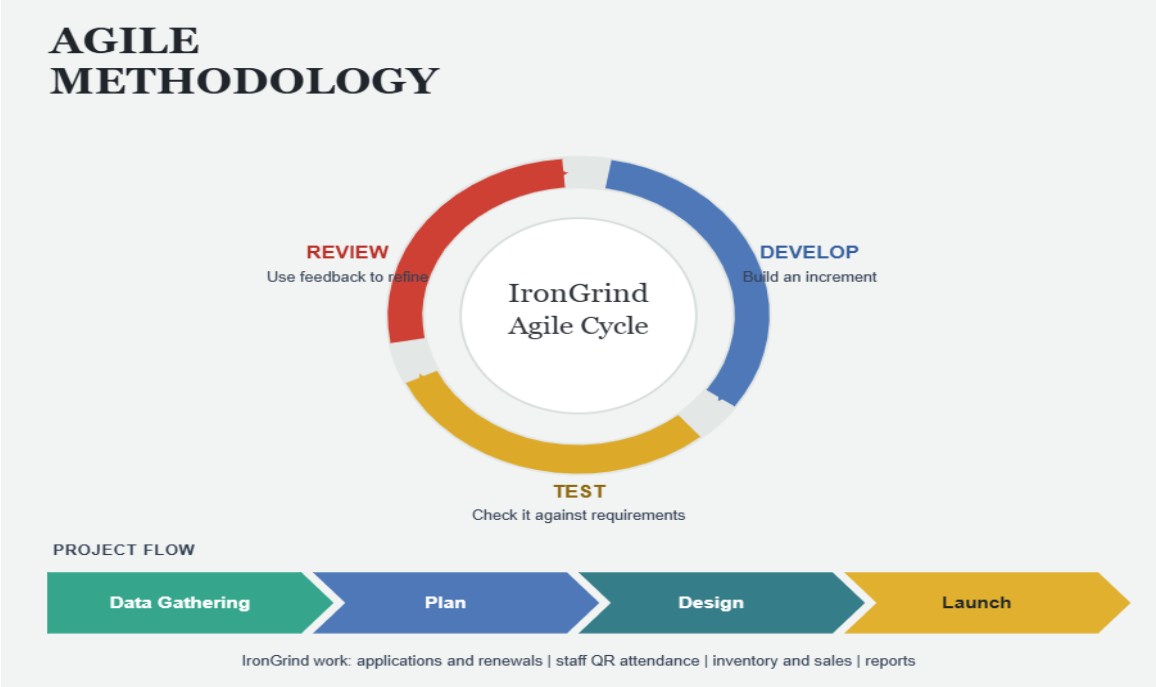


Figure 1. Agile

Agile is an iterative system-development approach in which stakeholder needs are gathered and prioritized, selected work is developed and tested in increments, and review results inform the next iteration. Figure 1 presents two connected flows for IronGrind. The circular cycle represents the repeated Develop, Test, and Review activities; feedback from a review informs the next increment. The lower project flow shows Data Gathering, Plan, and Design, followed by Launch only after the completed increments have been verified. The system work is organized around the gym's applicant/member and administrator workflows. The project does not assume a specific Agile framework such as Scrum; the research team should document the actual iteration length and review practices used and confirm that Agile is the methodology approved in the study proposal.

Requirements Planning

Requirements planning begins with gathering information about the gym's workflows and needs from the owner and authorized staff and reviewing the project scope in Chapters 1 and 2. The information is translated into system requirements and organized in a prioritized task list, beginning with account access and membership applications, followed by staff review, member access, attendance, inventory, sales, and reporting. Each selected requirement is given acceptance criteria. The final study should identify the data-gathering techniques and participants actually used. Privacy and authorization requirements apply throughout development because the system handles personal and health-related information.

Prototype

The system prototype is an interactive, browser-based version of the proposed gym workflows rather than a set of static screen mockups. Applicants can use the sign-in and registration pages to enter the application workflow. Members use the member portal to review membership information, display their QR code, view attendance history, and submit an eligible renewal request. Administrators use the dashboard to review applications and manage membership, staff-scanned attendance, products, sales, receipts, and reports. The prototype uses Supabase Auth and the configured database services for supported operations. Its key pages are listed in the User Interface Modules section. It should be treated as a prototype until deployment and operational readiness have been verified.

Construction

Selected requirements are implemented in the HTML, CSS, and JavaScript client and the Supabase authentication and PostgreSQL services. Database policies and functions support authorization and operational transactions. Each increment is tested against its acceptance criteria. At the end of an iteration, issues and feedback are recorded, priorities are updated, and unfinished or revised work returns to the prioritized task list. Stakeholder reviews or user feedback should be reported as completed only when the team has records of those activities.

Implementation

Launch takes place after the required increments pass verification; it is not repeated at the end of every iteration. The website and Supabase project can then be configured for the intended environment using the correct project settings, access policies, and authorized administrator accounts. Staff should verify the workflows and receive appropriate orientation. This chapter does

not claim a production deployment, completed training, or user acceptance unless those activities are documented separately.

Requirements Specification

System Requirements

Table 1. Functional Requirements

Table 2. Nonfunctional Requirements

Table 3. Software Requirements

Table 4. Hardware Requirements

Table 5. Inputs and Outputs

System Design

IRON GRIND FITNESS GYM MEMBERSHIP & OPERATIONS MANAGEMENT SYSTEM

SYSTEM ARCHITECTURE

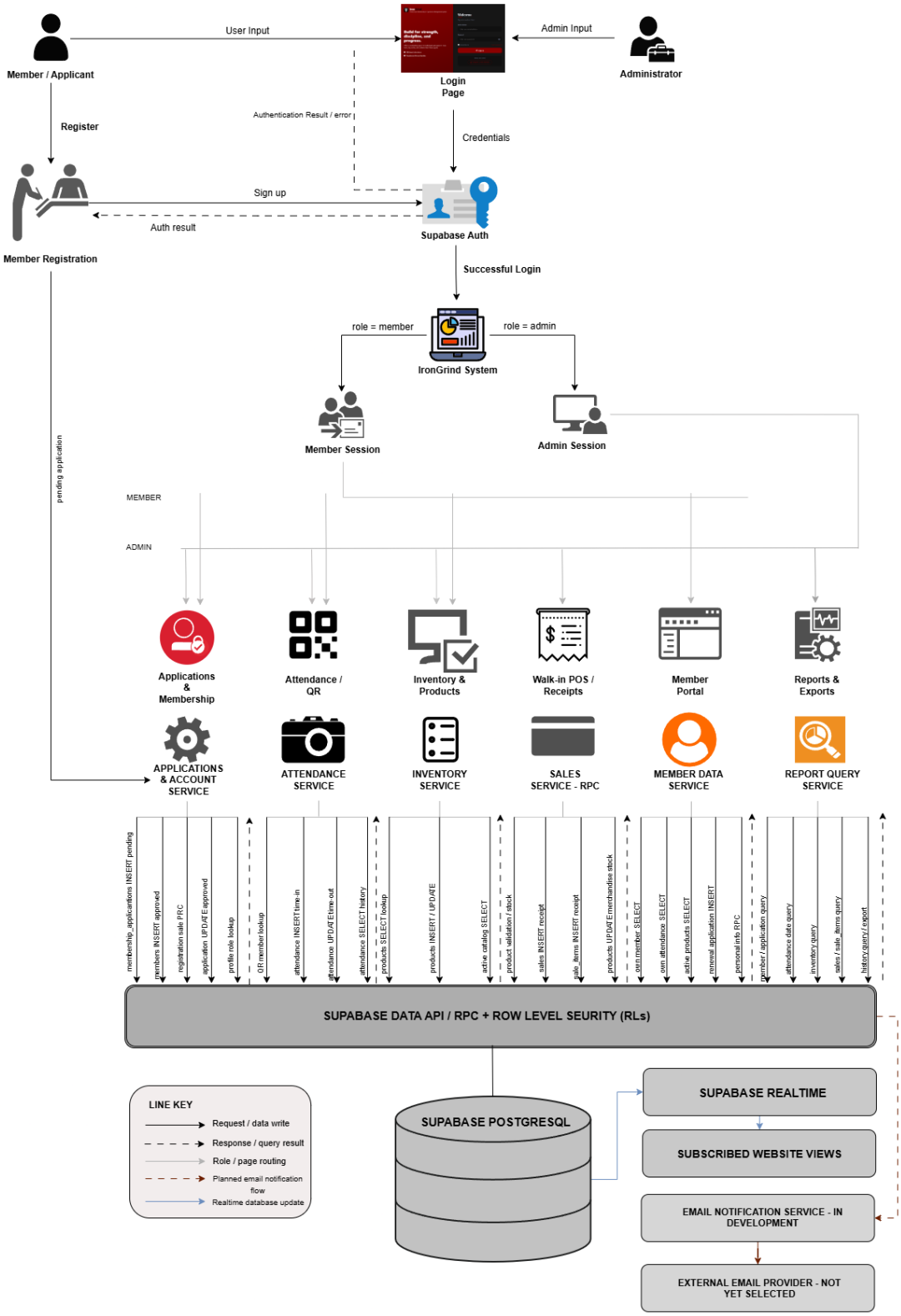


Figure 2. System Architecture Diagram

Figure 2 shows applicants and members using the login, registration, and member pages, while administrators use the admin dashboard. Supabase Auth verifies credentials and creates the authenticated session; the application then routes users to the member or administrator experience based on profile role. The browser modules are organized as Applications and Membership, Attendance and QR, Inventory and Products, Walk-in POS and Receipts, Member Portal, and Reports and Exports. These modules send requests through the Supabase Data API or database RPC functions. PostgreSQL Row Level Security (RLS) and function-level checks govern access to the Supabase PostgreSQL database, which stores profiles, membership applications, members, attendance, products, sales, sale items, and membership history.

The current Realtime configuration publishes changes from membership_applications and attendance to subscribed website views. The diagram also shows email notification delivery as in development, with no external email provider selected; it should not be described as an operational feature until it is implemented and configured. The Node.js server in the repository serves static files for local development and is not the production database backend.

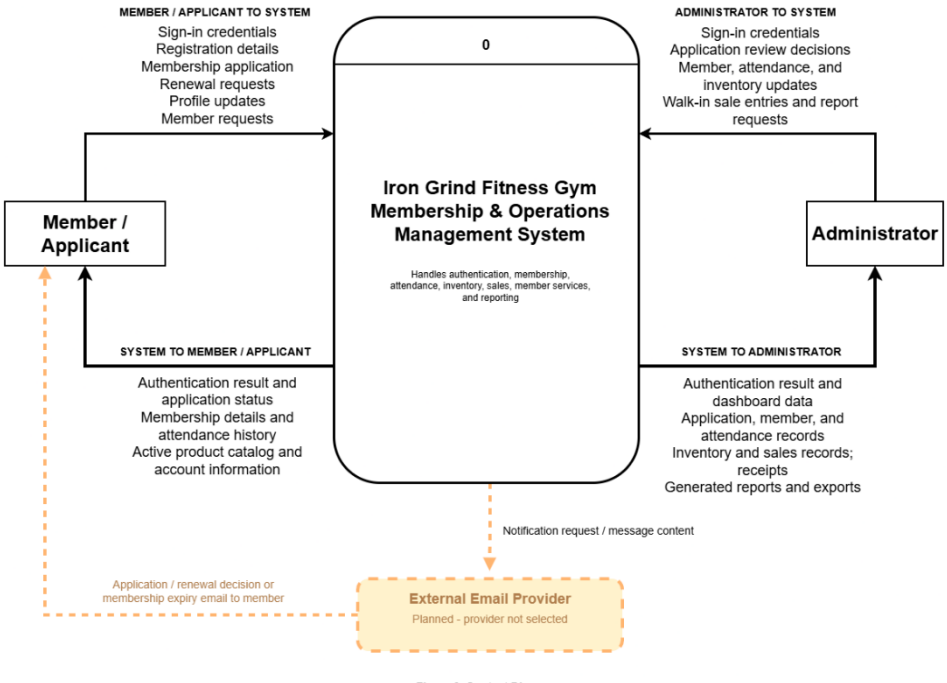


Figure 3. Context Diagram

The context diagram in Figure 3 illustrates the overall interaction between the external entities and the Iron Grind Fitness Gym Membership & Operations Management System. The central process serves as the main hub that manages authentication, membership, attendance, inventory, sales, member services, and reporting. The system interacts with two primary human actors: the Member/Applicant and the Administrator, along with a planned integration for an External Email Provider.

The Member/Applicant serves as the primary user who submits data to the system, including sign-in credentials, registration details, membership applications, renewal requests, and profile updates. In return, the system produces outputs for the member such as authentication results, application statuses, membership details, attendance history, and the active product catalog. Concurrently, the Administrator inputs operational data, including sign-in credentials, application review decisions, member and inventory updates, walk-in sale entries, and report requests. The system then produces outputs for the Administrator, including dashboard data, detailed member and attendance records, inventory and sales receipts, and generated reports.

Additionally, the diagram shows a planned interaction with an External Email Provider (represented by dashed lines). The system sends notification requests to this provider, which then sends application decisions and membership expiry emails directly to the members. Overall, the context diagram presents a clear overview of how the users interact with the system to maintain efficient, organized, and accurate gym membership and operational information management.